

LANDSCAPES OF THE CHURCH

03 — Green & Living Landscapes

Diocesan and Episcopal Conference layers: overview, ArcGIS Online-ready descriptions, methodology, QC, formulas, workflow, and reproducible code

LSAP Integral Ecology Atlas | Technical Documentation | 3 October 2026

8. Analysis 03 — Green & Living Landscapes

8.1 Diocesan layer overview

Layer name	LandscapesChurch_03_Dioceses_GreenLiving_V2
Layer alias	03 - Diocesan Green & Living Landscapes
Scale / features	Diocese; 2,874 features
Description	Combines trees and other vegetation to describe mapped tree/vegetation cover and its internal balance within each diocese. The layer does not claim to represent every living land system, because crops and rangelands are also biologically living systems.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Trees + other vegetation combined extent; tree/vegetation shares within green cover; leading component; balance/context; analytical ranks retained.
Key story fields	STORY_GREEN; STORY_CHARACTER; STORY_SCALE; STORY_CHURCH; STORY_SUMMARY; STORY_COMPREHENSIVE
Status	Publication-ready

8.2 Episcopal Conference layer overview

Layer name	LandscapesChurch_03_Conferences_GreenLiving_V2
Layer alias	03 - Episcopal Conference Green & Living Landscapes
Scale / features	Episcopal Conference; 122 features
Description	Summarizes mapped trees and other vegetation across Episcopal Conference geographies and relates those landscape systems to Church and LSAP presence.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).

Analyses	Trees + other vegetation combined extent; internal balance; Conference/Church/LSAP narrative context.
Key story fields	Diocesan story set plus STORY_LSAP.
Status	Publication-ready

8.3 Formulae and fields

Green share	$GREEN_PCT = PCT_LULC_Trees + PCT_LULC_Veg$ Mapped trees plus the source's other-vegetation class.
-------------	--

Green area	$GREEN_AREA = Trees_km^2 + OtherVegetation_km^2$ Area in km ² after source conversion.
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Tree share within green cover	$TREE_SHARE_GREEN = 100 \times Trees / (Trees + OtherVegetation)$ Internal composition of the combined green-cover system.
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Other vegetation share	$VEG_SHARE_GREEN = 100 \times OtherVegetation / (Trees + OtherVegetation)$ Complement of TREE_SHARE_GREEN.
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- GREEN_PRIMARY
- GREEN_BALANCE
- GREEN_CLASS
- TREE_RANK / TREE_PCTL
- VEG_RANK / VEG_PCTL

8.4 Core QC

Retained QC output

GREEN & LIVING LANDSCAPES – DISTRIBUTION / CONTENT QC

Dioceses:

Trees primary: 2,838 (98.7%)
Other vegetation: 36 (1.3%)

Episcopal Conferences:

Trees primary: 122 (100.0%)
Other vegetation: 0 (0.0%)

Scientific / narrative refinements:

- “Trees + Other Vegetation” is not described as all living terrestrial cover.
- Near-zero values render as <0.1%.
- Zero Catholic counts are not narrated as literal “0 Catholics.”
- Conference GREEN_CLASS labels were removed from story prose while retained analytically.
- Final formula/QC error count: 0.

Relevant reproducible code

A.6 Analysis 03 — Green & Living Landscapes

```
trees = safe(row["PCT_LULC_Trees"])
veg = safe(row["PCT_LULC_Veg"])

green_pct = trees + veg
```

```

if green_pct > 0:
    tree_share_green = 100.0 * trees / green_pct
    veg_share_green = 100.0 * veg / green_pct
else:
    tree_share_green = 0.0
    veg_share_green = 0.0

green_primary = (
    "Trees"
    if trees >= veg
    else "Other Vegetation"
)

green_area = (
    safe(row["TOTAL_LULC_Trees"]) +
    safe(row["TOTAL_LULC_Veg"])
)

```

A.10 Safe m² → km² conversion

```

import statistics

values = []

with arcpy.da.SearchCursor(
    feature_class,
    ["ANALYSIS_AREA"]
) as cursor:

    for (value,) in cursor:
        if value is not None and value > 0:
            values.append(float(value))

median_area = statistics.median(values)

if median_area >= 1_000_000:

    with arcpy.da.UpdateCursor(
        feature_class,
        ["ANALYSIS_AREA"]
    ) as cursor:

        for row in cursor:
            if row[0] is not None:
                row[0] = float(row[0]) / 1_000_000.0
                cursor.updateRow(row)

else:
    print("Area already appears to be in km²; no conversion performed.")

```

A.11 Rank / percentile reference implementation

```

def competition_rank(values):
    """
    values = iterable of (feature_id, numeric_value)
    Rank 1 = highest.
    Ties receive the same rank; the next rank skips accordingly.
    """
    ordered = sorted(
        values,
        key=lambda x: x[1],
        reverse=True
    )

    result = {}
    previous = None
    current_rank = 0

    for position, (feature_id, value) in enumerate(ordered, start=1):
        if previous is None or value != previous:
            current_rank = position

```

```

        result[feature_id] = current_rank
        previous = value

    return result

def percentile_at_or_below(all_values, value):
    valid = [v for v in all_values if v is not None]

    if not valid:
        return None

    n = sum(v <= value for v in valid)

    return 100.0 * n / len(valid)

```

A.16 Publication cleanup pattern

```

# Keep only fields intentionally required for the publication derivative.
KEEP_FIELDS = {
    "BUFFER_ID",
    "Account_Name",
    "Country",
    "Language",
    "Official_website",
    "Sector",
    "BUFFER_AREA",
    "LANDSCAPE_SCALE",
    # ... eight PCT_LULC fields ...
    # ... coverage/mosaic/transition fields ...
    # ... story fields ...
}

delete_fields = [
    f.name
    for f in arcpy.ListFields(feature_class)
    if (
        not f.required and
        f.name not in KEEP_FIELDS
    )
]

if delete_fields:
    arcpy.management.DeleteField(
        feature_class,
        delete_fields
    )

# Always run a final required-field check after deletion.

```

Sources and attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas / project Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and the underlying credited source organizations.

Esri Global Biodiversity and Conservation Hexagons: <https://www.arcgis.com/home/item.html?id=f1bdbcac3c54ad8b5b21f5b17d91e8d>

Esri 2026 update: <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>

Summarize Within documentation: <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>